

DIGITAL FORENSICS LABORATORY REPORT

Module 7: Data Carving and File Recovery Techniques

Evidence Item: Ch01InChap01.dd

Evidence File	Ch01InChap01.dd
Analysis Platform	Kali Linux (VirtualBox VM)
Tools Used	The Sleuth Kit (fls, fsstat, icat), md5sum, sha256sum, strings
Report Date	September 5, 2026

1. Executive Summary

This report documents a forensic examination of the raw volume image Ch01InChap01.dd, conducted as part of Module 7: Data Carving and File Recovery Techniques. The objective of the examination was to establish and preserve the evidentiary integrity of the image, analyze its underlying file system structure, and recover files that had been marked as deleted but remained physically present on the media.

The examination was performed using The Sleuth Kit (TSK) suite of command-line forensic utilities on a Kali Linux analysis workstation. The image was identified as a FAT12 volume containing a mixture of allocated and deleted files. Three previously deleted files — a billing letter, a regrets letter, and a confirmation notice — were successfully recovered using the icat data-unit extraction tool.

Cryptographic hashing was performed at each stage of the process to verify that the extraction did not alter the underlying evidence.

2. Evidence Management and Chain of Custody

Prior to analysis, a dedicated working directory (~/forensic_lab) was created and the evidence file was transferred into it, isolating the working copy from the original acquisition location. This preserves a clear separation between the source evidence and the analyst's working environment.

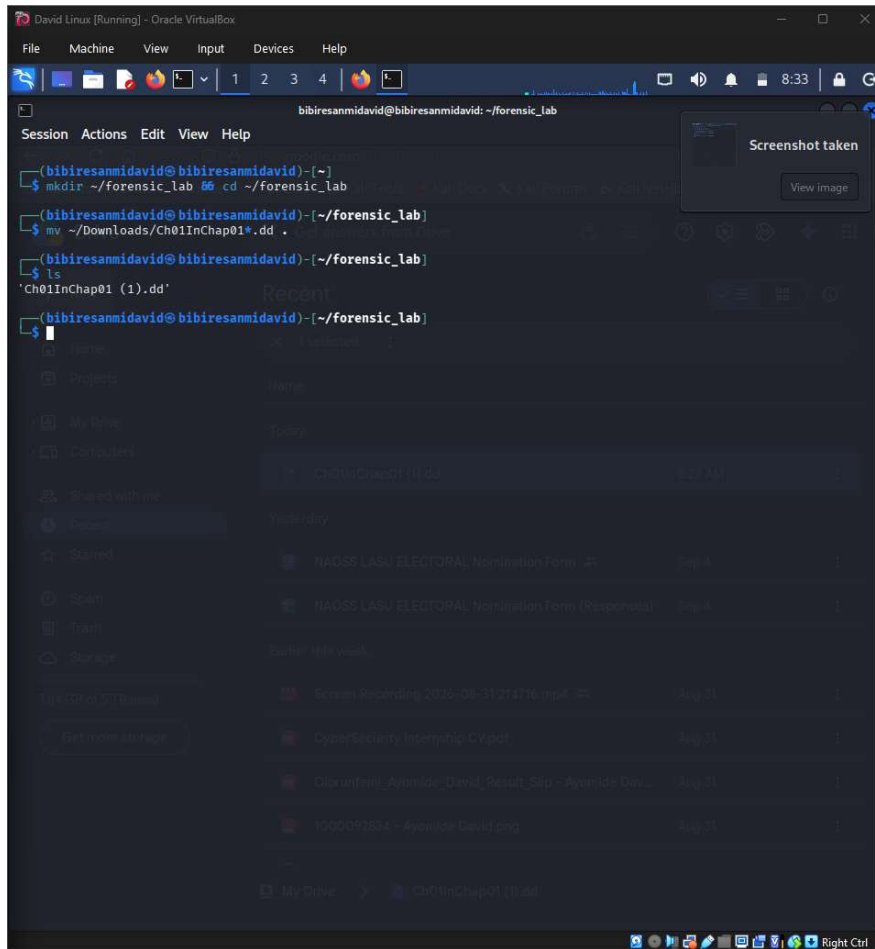


Figure 1. Evidence preparation — working directory creation and transfer of Ch01InChap01.dd.

Immediately upon receipt, baseline cryptographic hash values were computed for the evidence file to establish a fixed, verifiable reference point for the state of the evidence at the start of the examination. Both MD5 and SHA-256 algorithms were used to provide cross-validated integrity assurance.

Evidence File	Algorithm	Hash Value
Ch01InChap01.dd	MD5	a117773bcf1fc88ec0ab8e0a349fbbcb
Ch01InChap01.dd	SHA-256	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

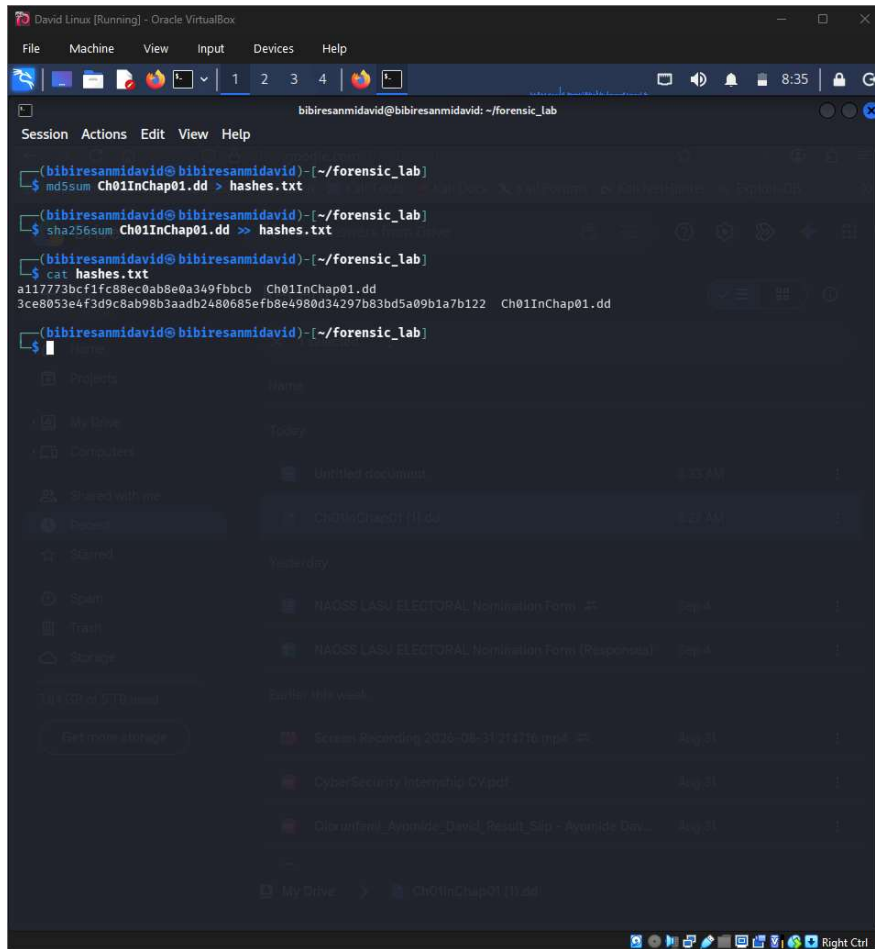


Figure 2. Baseline MD5 and SHA-256 hashes computed for Ch01InChap01.dd.

3. File System and Partition Analysis

Analysis of the volume's structure and contents was conducted using two TSK utilities: fls, to enumerate the file and directory entries recorded within the file system, and fsstat, to extract detailed file system metadata.

3.1 File Listing (fls)

The fls utility was run directly against the image to list all file entries, including those that remained allocated as well as entries flagged as deleted (denoted with an asterisk). This listing distinguishes recoverable content from files still present in the active file system.

Inode	File Name	Status
5	Client Info.mdb	Allocated
8	Billing Letter.doc	Deleted
11	confirmation.txt	Deleted

13	Income.xls	Allocated
15	letter1.txt	Deleted
17	Regrets.doc	Deleted

```

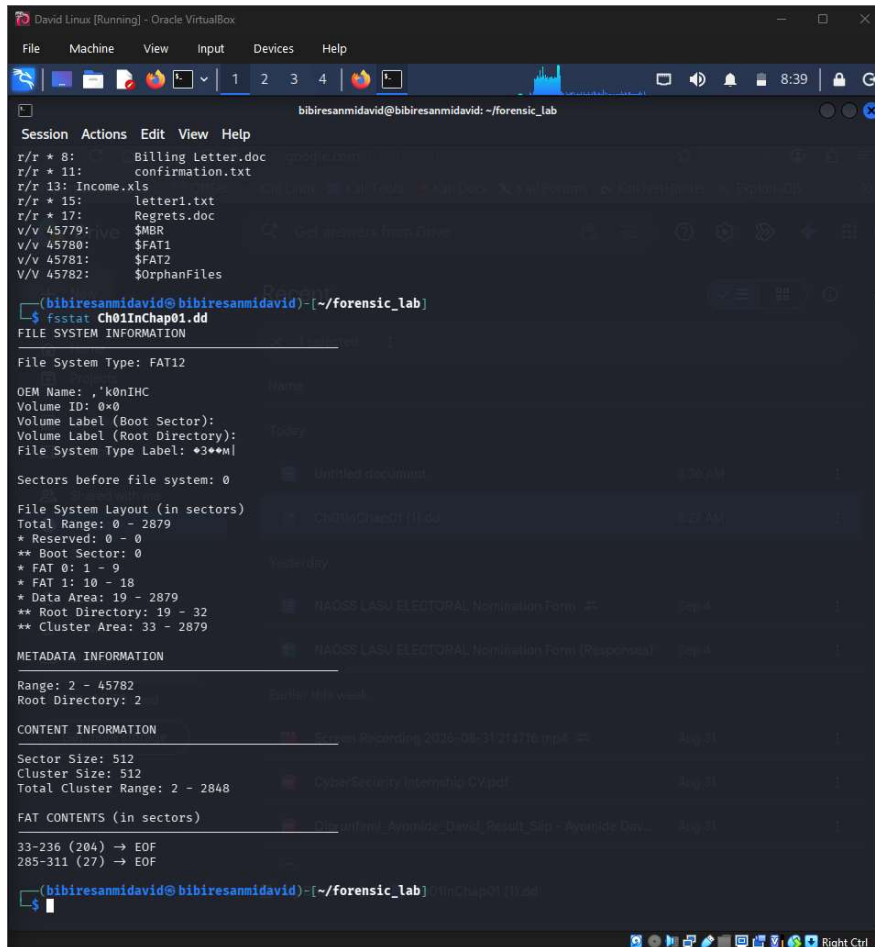
(bibiresanmidavid@bibiresanmidavid)~[/forensic_lab]
$ md5sum Ch01InChap01.dd > hashes.txt
a117773b0ef1f288ec0ab8e0a349fbbcb Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd
(bibiresanmidavid@bibiresanmidavid)~[/forensic_lab]
$ mmls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
v/v 45782: $OrphanFiles
(bibiresanmidavid@bibiresanmidavid)~[/forensic_lab]
$ tsk fs

```

Figure 3. TSK fs output listing allocated and deleted file entries within the volume.

3.2 File System Statistics (fsstat)

The fsstat utility was used to extract the file system metadata directly, without requiring a separate partition-mapping step (see Section 6, Limitations). The output confirmed the volume as a FAT12 file system with a sector size of 512 bytes and a cluster size of 512 bytes, and provided the metadata address ranges used to correlate inode numbers with their corresponding data units.



```
David Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

bibiresanmidavid@bibiresanmidavid: ~/forensic_lab

Session Actions Edit View Help
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
v/v 45782: $OrphanFiles

(bibiresanmidavid@bibiresanmidavid)~[~/forensic_lab]
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
File System Type: FAT12
OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: *3**m|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)
33-236 (204) -> EOF
285-311 (27) -> EOF

(bibiresanmidavid@bibiresanmidavid)~[~/forensic_lab] Ch01InChap01 (1).dd
```

Figure 4. TSK fsstat output detailing FAT12 file system layout and metadata ranges.

4. Evidence Extraction and Verification

The icat utility was used to extract the content of each deleted file directly from its data units by inode number, bypassing the file system's (now-cleared) directory entry. Three files were recovered in this manner: Billing Letter.doc (inode 8), Regrets.doc (inode 17), and confirmation.txt (inode 11).

Following extraction of the billing letter, the strings utility was used to preview its readable text content, confirming that the recovered data was coherent, human-readable business correspondence rather than corrupted or partial fragments.

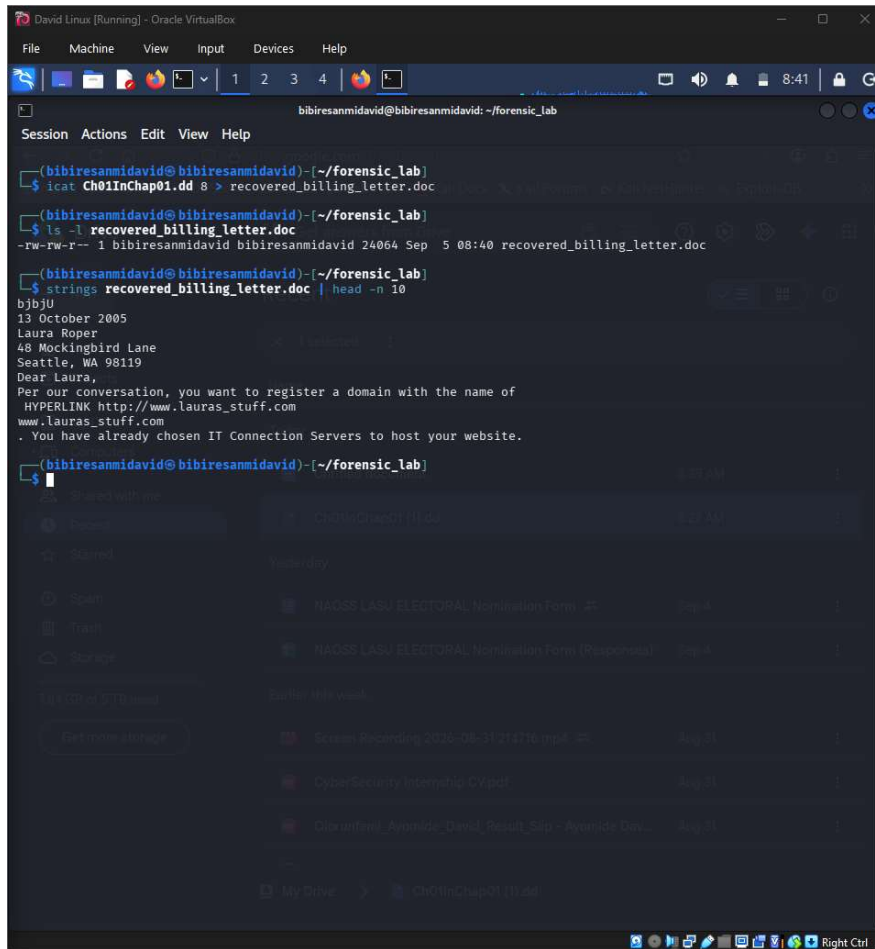
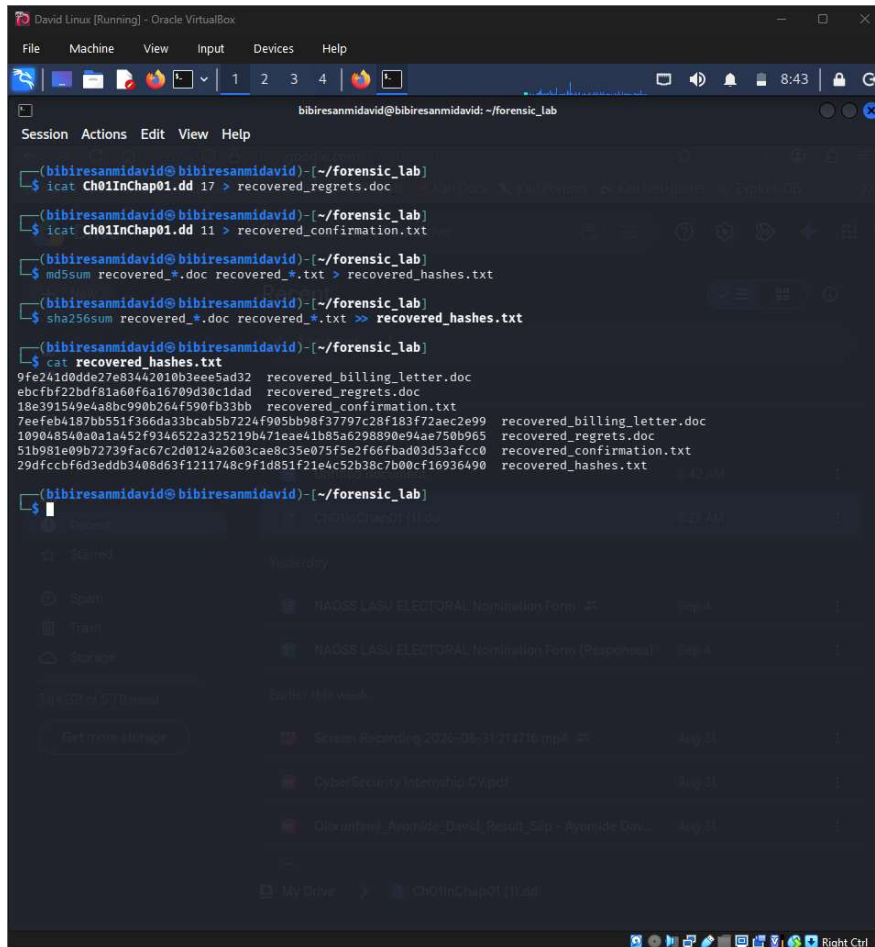


Figure 5. icat extraction of the billing letter and a strings preview confirming readable content.

The remaining two deleted files (Regrets.doc and confirmation.txt) were then extracted using the same icat procedure. To verify the integrity of the extraction process, MD5 and SHA-256 hashes were computed for all three recovered files and recorded in recovered_hashes.txt.

Recovered File	MD5	SHA-256
recovered_billing_letter.doc	9fe241d0dde27e83442010b3eee5ad32	7eefeb4187bb551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99
recovered_regrets.doc	ebcfbf22bdf81a60f6a16709d30c1dad	109048540a0a1a452f9346522a3252196471eae41b85a6298890e94ae750b965
recovered_confirmation.txt	18e391549e4a8bc990b264f590fb33bb	51b981e09b72739fac67c2d0124a2603cae8c35e075f5e2f66fbad03d53afcc0



```
(bibiresanmidavid@bibiresanmidavid) [~/forensic_lab]
$ icat Ch01InChap01.dd 17 > recovered_regrets.doc
(bibiresanmidavid@bibiresanmidavid) [~/forensic_lab]
$ icat Ch01InChap01.dd 11 > recovered_confirmation.txt
(bibiresanmidavid@bibiresanmidavid) [~/forensic_lab]
$ md5sum recovered_*.doc recovered_*.txt > recovered_hashes.txt
(bibiresanmidavid@bibiresanmidavid) [~/forensic_lab]
$ sha256sum recovered_*.doc recovered_*.txt >> recovered_hashes.txt
(bibiresanmidavid@bibiresanmidavid) [~/forensic_lab]
$ cat recovered_hashes.txt
9fe241d0de27e83442010b3ee5ad32 recovered_billing_letter.doc
ebcfbf22bdf81a0f6a16709d30c1dad recovered_regrets.doc
18e391549ea8bc990b264f598fb3bb recovered_confirmation.txt
7eeFeb4187b551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99 recovered_billing_letter.doc
109048540a01a452f9346522a325219b471eae41b85a6298890e94ae750b965 recovered_regrets.doc
51b981e09b72739fac67c2d0124a2603cae8c35e075f5e2f66fbad03d53afcc0 recovered_confirmation.txt
29dfccbf6d3eddb3408d63f1211748c9f1d851f21e4c52b38c7b00cf16936490 recovered_hashes.txt
(bibiresanmidavid@bibiresanmidavid) [~/forensic_lab]
$
```

Figure 6. Recovered file extraction and computation of verification hashes (recovered_hashes.txt).

5. Findings and Analysis

- The evidence image Ch01InChap01.dd contains a FAT12 file system holding a mixture of active and deleted business records.
- Three deleted files were successfully recovered in full using inode-based data-unit extraction (icat), indicating that their underlying clusters had not yet been overwritten by subsequent write activity.
- The recovered billing letter contains business correspondence referencing a domain name registration for client Laura Roper (48 Mockingbird Lane, Seattle, WA), specifying the domain `www.lauras_stuff.com` and IT Connection Servers as the intended hosting provider.
- Independent MD5 and SHA-256 hashing of the evidence file and of each recovered file provides a verifiable record that supports the integrity of the examination and the authenticity of the recovered content.

Taken together, these findings demonstrate that meaningful evidentiary content can be recovered from a FAT file system after file deletion, provided the underlying data units have not been reallocated, and that this recovery can be performed and verified without altering the source evidence.

6. Limitations

The evidence item Ch01InChap01.dd is a raw, unpartitioned volume image rather than a full disk image containing a partition table. Consequently, the mmls utility — normally used to map partition layout prior to file system analysis — returned no partition information and was not applicable to this examination. File system analysis was instead performed directly at the volume level using fls and fsstat, which is the correct and expected approach for an unpartitioned volume image of this type.

This examination was limited to file recovery via direct inode/data-unit extraction. Unallocated space was not carved for file fragments outside of the three identified deleted entries, and no timeline or metadata-timestamp analysis was performed as part of this module.